

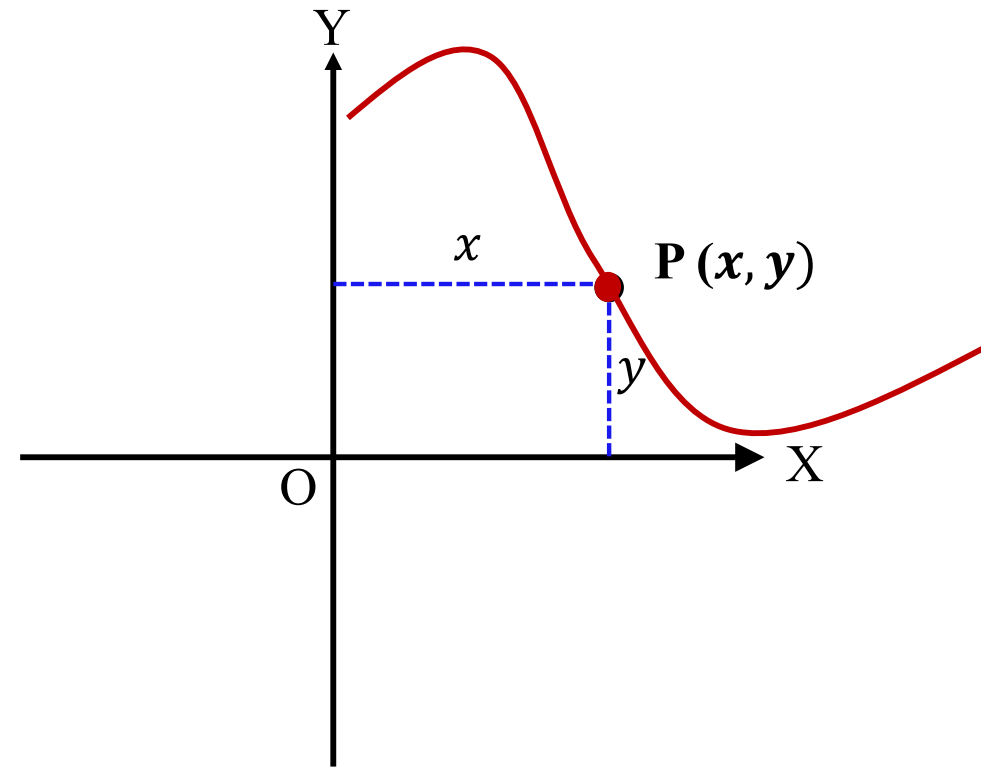
Lecture 03

Coordinate systems



Cartesian coordinate system in plane

- Instantaneous position P can be represented by measuring its perpendicular distances (x and y) from two perpendicularly intersecting lines (called Axes) passing through the origin.
- In Cartesian coordinate, the position P is represented by (x, y) .

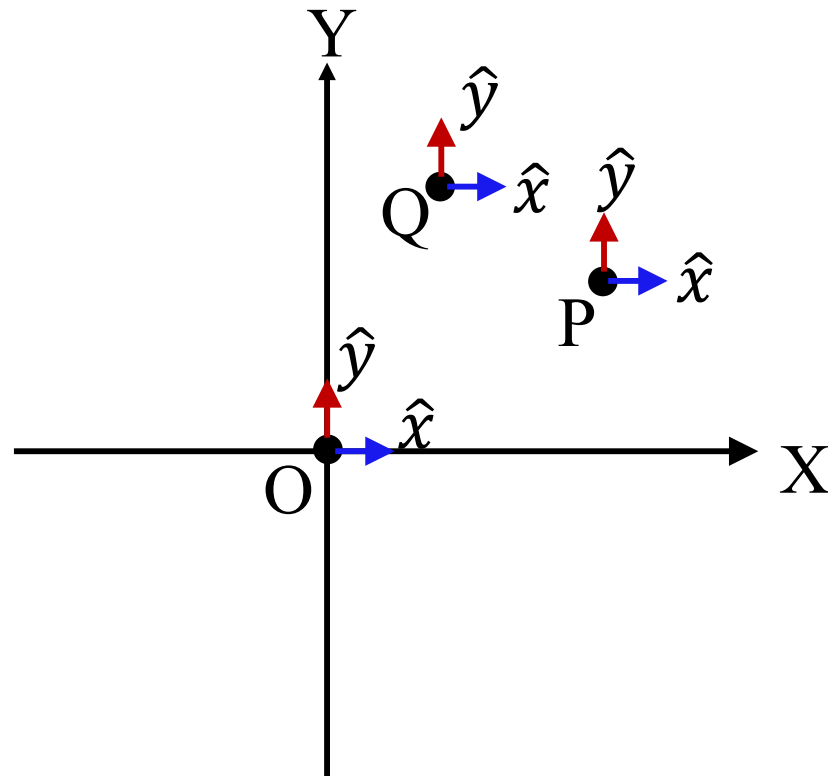


Cartesian coordinate system in plane

All points in the plane are therefore referred with respect to same set of X and Y.



Unit vectors in the plane: Cartesian



- In Cartesian coordinate system, one can define two unit vectors $\hat{x}$ and $\hat{y}$ at every point in the plane.
- $\hat{x}$ and $\hat{y}$ are unit vectors in the increasing direction of x and y , thus they are parallel to coordinate axes X and Y , respectively.
- $\hat{x}$ and $\hat{y}$ are orthogonal, and directed in the same direction at every points.

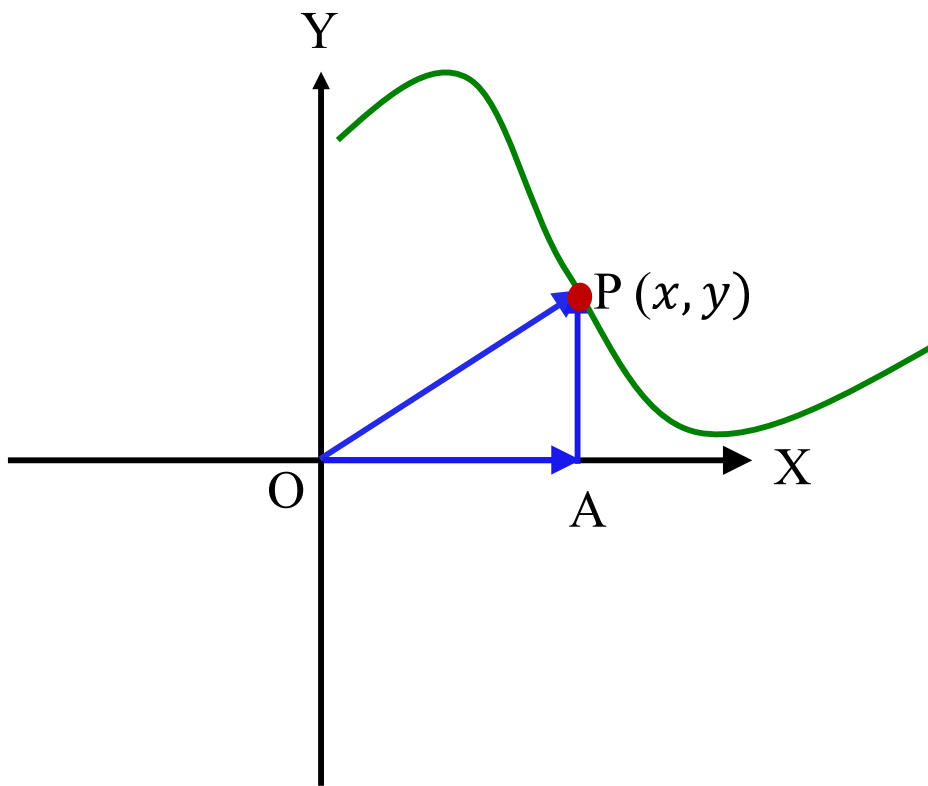
Another way of looking unit vector Cartesian coordinate in plane

$\hat{x}$ is the unit vector perpendicular to $x = \text{constant}$ line

$\hat{y}$ is the unit vector perpendicular to $y = \text{constant}$ line



Position vector in terms of Cartesian components



- Position **P** can be represented through Cartesian coordinate (x, y) .
 - From vector addition rule
$$\overrightarrow{OP} = \overrightarrow{OA} + \overrightarrow{AP}$$
- $\overrightarrow{OA} =$ (magnitude of OA) (unit vector along/parallel to X-direction) $= x\hat{x}$
- $\overrightarrow{AP} =$ (magnitude of AP) (unit vector along/parallel to Y-direction) $= y\hat{y}$

We may also use the notation:

$$\begin{aligned}\hat{x} &= \hat{i} \\ \hat{y} &= \hat{j}\end{aligned}$$

$$\text{Thus, } \overrightarrow{OP} = \vec{r} = x\hat{x} + y\hat{y}$$



Velocity and acceleration in Cartesian

Velocity in Cartesian coordinate

$$\vec{v} = \frac{d\vec{r}}{dt} = \frac{d}{dt} (x \hat{x} + y \hat{y})$$

$$\Rightarrow = \dot{x} \hat{x} + x \frac{d\hat{x}}{dt} + \dot{y} \hat{y} + y \frac{d\hat{y}}{dt}$$

$$\Rightarrow \vec{v} = \dot{x} \hat{x} + \dot{y} \hat{y}$$

$$\begin{aligned} \frac{d\hat{x}}{dt} &= 0 \\ \frac{d\hat{y}}{dt} &= 0 \end{aligned}$$

**Standard
Notations:**

$$\frac{dx}{dt} = \dot{x}$$

Acceleration $\vec{a} = \frac{d\vec{v}}{dt} = \ddot{x} \hat{x} + \ddot{y} \hat{y}$

Newton's second law in vector form

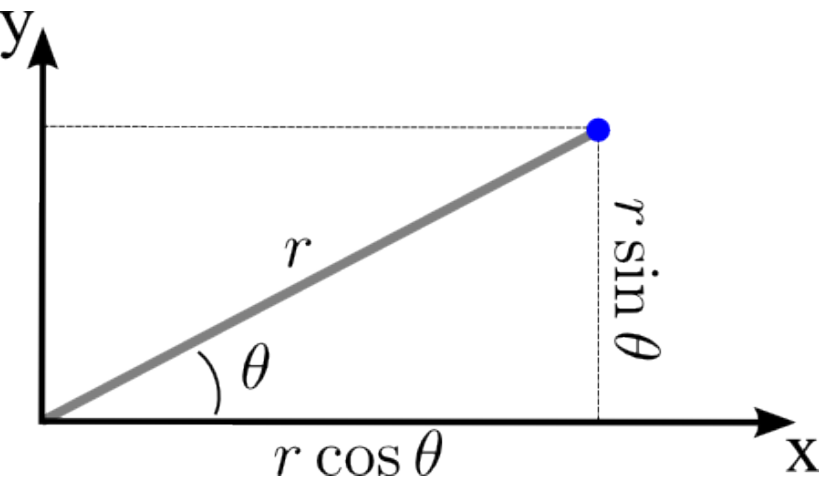
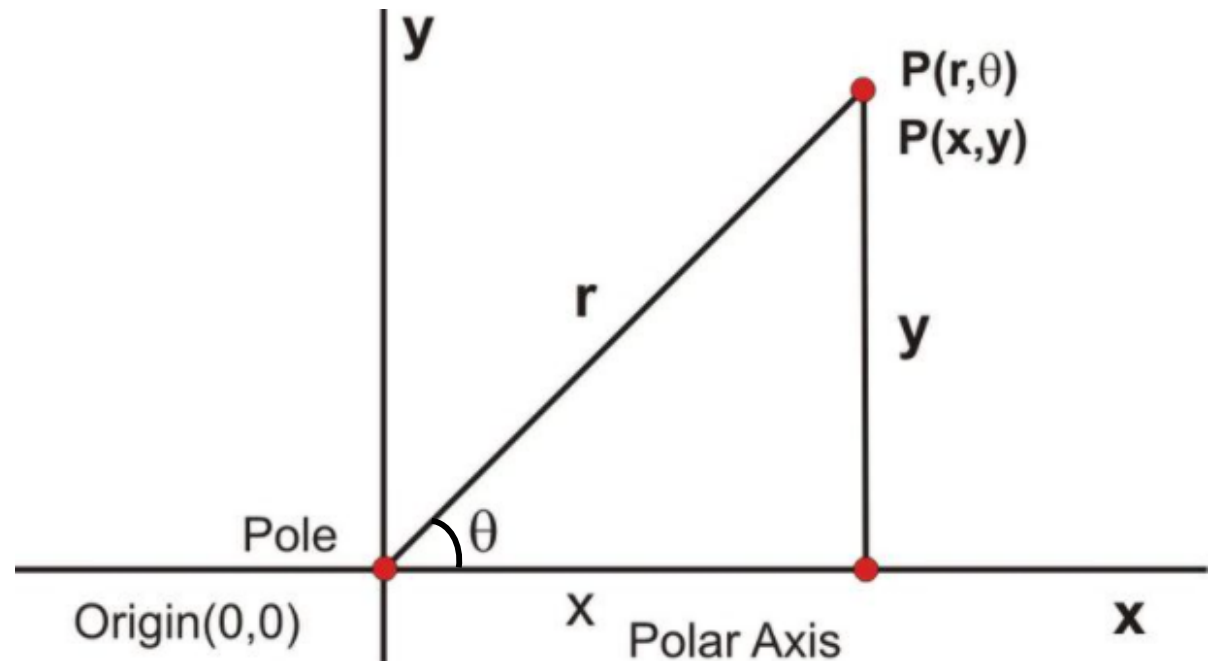
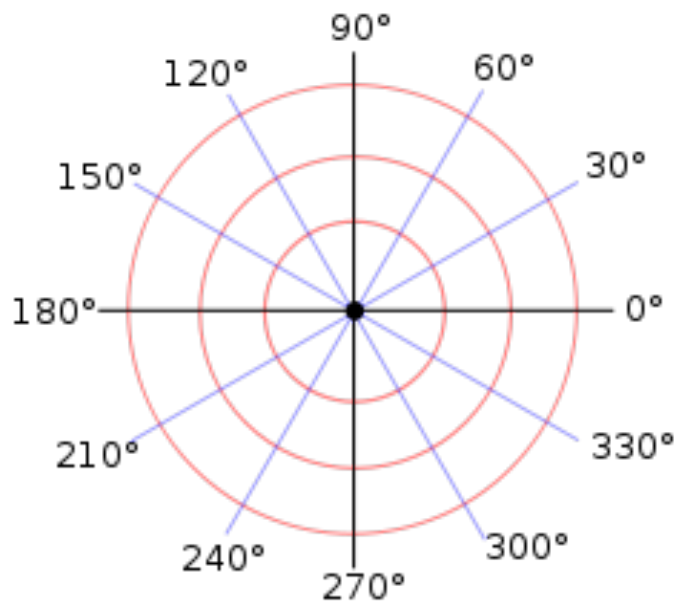
$$\vec{F} = F_x \hat{x} + F_y \hat{y} = m \frac{d\vec{v}}{dt} = m(\ddot{x} \hat{x} + \ddot{y} \hat{y})$$



Plane Polar Coordinates



Plane polar coordinates



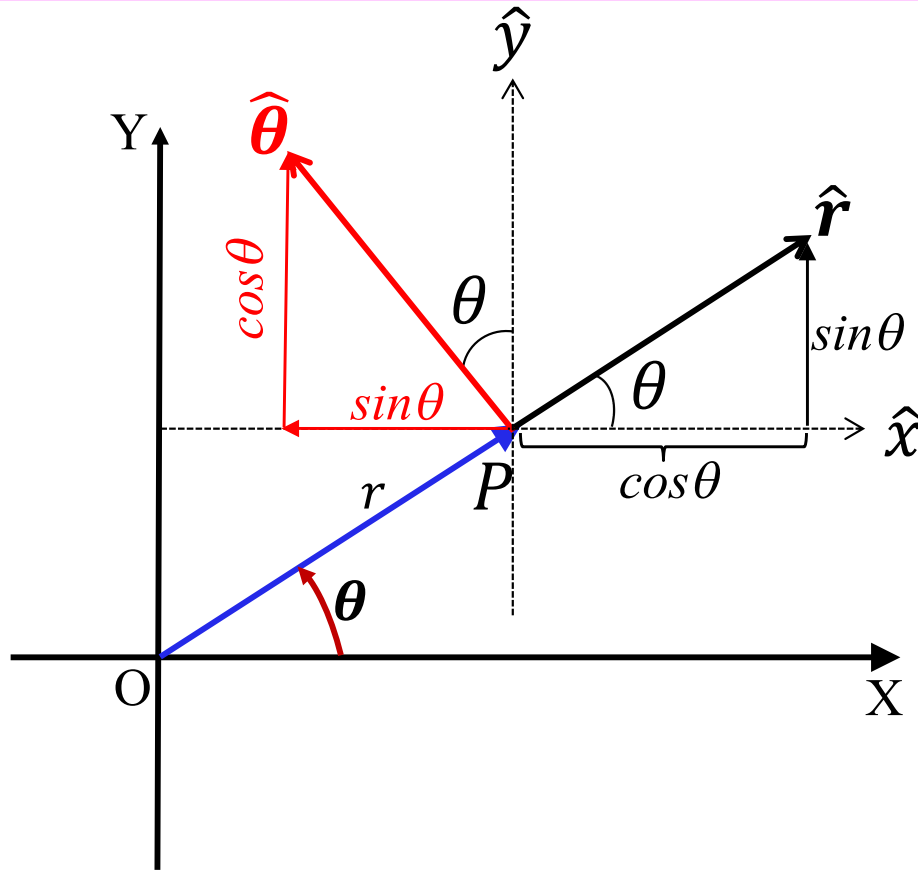
Each point $P(x, y)$ on the plane can also be represented by its distance, r from the origin O and the angle, θ , OP makes with X-axis.

→ Relationship with Cartesian coordinates

$$x = r \cos \theta \quad \& \quad y = r \sin \theta$$

$$r = (x^2 + y^2)^{1/2} \quad \& \quad \theta = \tan^{-1} \left(\frac{y}{x} \right)$$

Unit vector in plane polar coordinate



⊙ The unit vectors for plan polar coordinates are $\hat{r}$ and $\hat{\theta}$.

⊙ $\hat{r}$ and $\hat{\theta}$ are unit vectors along increasing direction of coordinate r and θ .

⊙ The direction of $\hat{r}$ and $\hat{\theta}$ varies with position.

$$\hat{r} = \cos \theta \hat{x} + \sin \theta \hat{y}$$

$$\hat{\theta} = -\sin \theta \hat{x} + \cos \theta \hat{y}$$

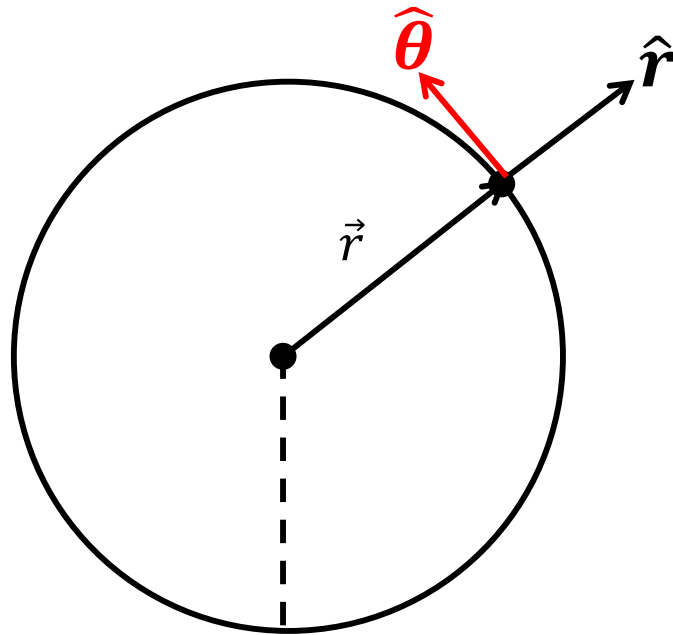
Unit vectors depend only on θ .

$\hat{r}$ and $\hat{\theta}$ are **orthogonal** $\hat{r} \cdot \hat{\theta} = 0$

Their directions depend on location.



Unit vector in plane polar coordinate



- ⊙ The position vector $\vec{r}$ in plane polar coordinates is given by $\vec{r} = r\hat{r}$.
- ⊙ In cartesian coordinates
$$\vec{r} = x\hat{x} + y\hat{y},$$

where $x = r \cos \theta$ & $y = r \sin \theta$.
- ⊙ Thus, $\vec{r} = r \cos \theta \hat{x} + r \sin \theta \hat{y}$

The unit vectors are defined as $\hat{r} = \left(\frac{\partial \vec{r} / \partial r}{|\partial \vec{r} / \partial r|} \right)$ and $\hat{\theta} = \left(\frac{\partial \vec{r} / \partial \theta}{|\partial \vec{r} / \partial \theta|} \right)$.

$$\hat{r} = \cos \theta \hat{x} + \sin \theta \hat{y}$$
$$\hat{\theta} = -\sin \theta \hat{x} + \cos \theta \hat{y}$$

Unit vectors' relation ($\hat{r}$, $\hat{\theta}$ vs $\hat{x}$, $\hat{y}$)

$$\begin{aligned}\hat{r} &= \cos \theta \hat{x} + \sin \theta \hat{y} \\ \hat{\theta} &= -\sin \theta \hat{x} + \cos \theta \hat{y}\end{aligned}$$

Multiply Eq. for $\hat{r}$ with $\cos \theta \rightarrow \cos \theta \hat{r} = \cos^2 \theta \hat{x} + \cos \theta \sin \theta \hat{y}$

Multiply Eq. for $\hat{\theta}$ with $\sin \theta \rightarrow \sin \theta \hat{\theta} = -\sin^2 \theta \hat{x} + \sin \theta \cos \theta \hat{y}$

Subtracting $\rightarrow$

$$\hat{x} = \cos \theta \hat{r} - \sin \theta \hat{\theta}$$

Multiply Eq. for $\hat{r}$ with $\sin \theta \rightarrow \sin \theta \hat{r} = \cos \theta \sin \theta \hat{x} + \sin^2 \theta \hat{y}$

Multiply Eq. for $\hat{\theta}$ with $\cos \theta \rightarrow \cos \theta \hat{\theta} = -\sin \theta \cos \theta \hat{x} + \cos^2 \theta \hat{y}$

Adding $\rightarrow$

$$\hat{y} = \sin \theta \hat{r} + \cos \theta \hat{\theta}$$

$$\hat{x} = \cos \theta \hat{r} - \sin \theta \hat{\theta}$$

$$\hat{y} = \sin \theta \hat{r} + \cos \theta \hat{\theta}$$



Derivative of Unit vectors in plane polar

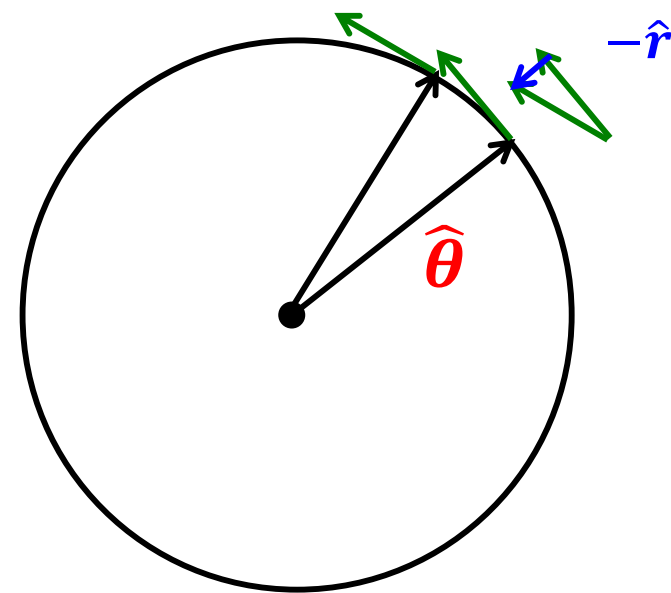
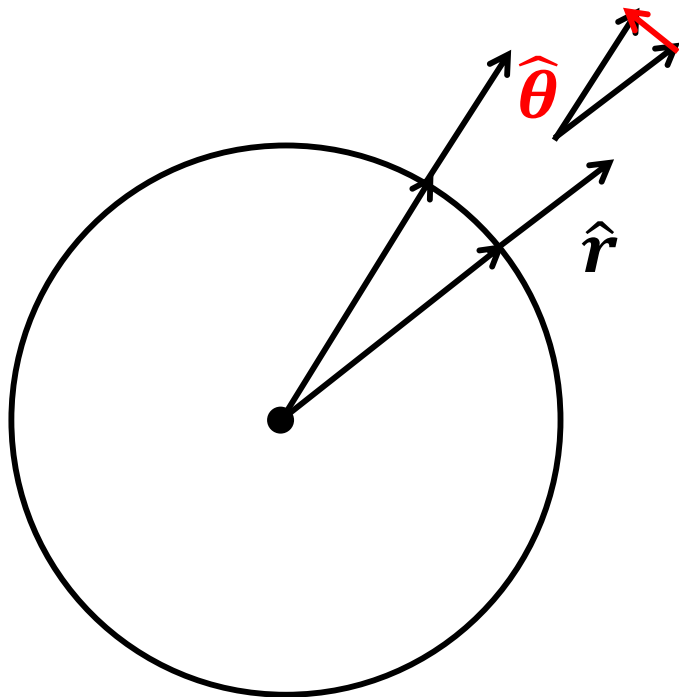
The unit vectors are function of θ only.

$$\hat{r} = \cos \theta \hat{x} + \sin \theta \hat{y}$$

$$\hat{\theta} = -\sin \theta \hat{x} + \cos \theta \hat{y}$$

$$\frac{\partial \hat{\theta}}{\partial \theta} = -\cos \theta \hat{x} - \sin \theta \hat{y} = -(\cos \theta \hat{x} + \sin \theta \hat{y}) = -\hat{r}$$

$$\frac{\partial \hat{r}}{\partial \theta} = -\sin \theta \hat{x} + \cos \theta \hat{y} = \hat{\theta}$$



Velocity in plane polar coordinate

Velocity $\vec{v} = \frac{d\vec{r}}{dt}$ $\vec{v} = \frac{d\vec{r}}{dt} = \frac{d(r\hat{r})}{dt}$

**Standard
Notations:**

$$\frac{dr}{dt} = \dot{r}$$

$$\frac{d^2\theta}{dt^2} = \ddot{\theta}$$

$$= \frac{dr}{dt} \hat{r} + r \frac{d\hat{r}}{dt}$$

$$= \dot{r} \hat{r} + r \frac{\partial \hat{r}}{\partial \theta} \frac{d\theta}{dt}$$

$$\boxed{\vec{v} = \dot{r} \hat{r} + r \dot{\theta} \hat{\theta}}$$

Since,

$$\frac{\partial \hat{r}}{\partial \theta} = \hat{\theta}$$

$$\frac{\partial \hat{\theta}}{\partial \theta} = -\hat{r}$$

Radial component, $\dot{r}$ and tangential/transverse component, $r\dot{\theta}$



Acceleration in plane polar coordinate

$$\vec{a} = \frac{d\vec{v}}{dt}$$

Note:

$$\frac{\partial \hat{r}}{\partial \theta} = \hat{\theta} \quad \& \quad \frac{\partial \hat{\theta}}{\partial \theta} = -\hat{r}$$

$$\begin{aligned} \frac{d\vec{v}}{dt} &= \frac{d}{dt} (\dot{r}\hat{r} + r\dot{\theta}\hat{\theta}) = \frac{dr}{dt}\hat{r} + \dot{r}\frac{d\hat{r}}{dt} + \frac{dr}{dt}\dot{\theta}\hat{\theta} + r\ddot{\theta}\hat{\theta} + r\dot{\theta}\frac{d\hat{\theta}}{dt} \\ &= \frac{dr}{dt}\hat{r} + \dot{r}\frac{d\hat{r}}{dt} + \dot{r}\dot{\theta}\hat{\theta} + r\ddot{\theta}\hat{\theta} + r\dot{\theta}\frac{d\hat{\theta}}{dt} \\ &= \dot{r}\hat{r} + \dot{r}\frac{\partial \hat{r}}{\partial \theta}\frac{d\theta}{dt} + \dot{r}\dot{\theta}\hat{\theta} + r\ddot{\theta}\hat{\theta} + r\dot{\theta}\frac{\partial \hat{\theta}}{\partial \theta}\frac{d\theta}{dt} \\ &= \dot{r}\hat{r} + \dot{r}\dot{\theta}\hat{\theta} + \dot{r}\dot{\theta}\hat{\theta} + r\ddot{\theta}\hat{\theta} - r\dot{\theta}^2\hat{r} \end{aligned}$$

$$\vec{a} = (\ddot{r} - r\dot{\theta}^2)\hat{r} + (2\dot{r}\dot{\theta} + r\ddot{\theta})\hat{\theta}$$



Acceleration in plane polar coordinate

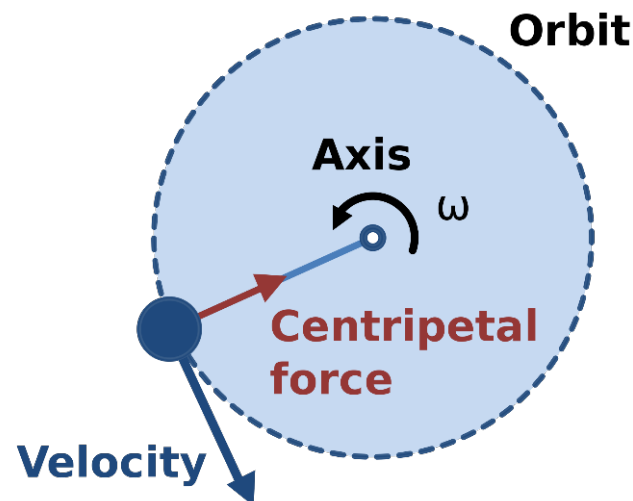
$$\vec{a} = (\ddot{r} - r\dot{\theta}^2)\hat{r} + (2\dot{r}\dot{\theta} + r\ddot{\theta})\hat{\theta}$$

Radial component $\ddot{r} - r\dot{\theta}^2$

($-r\dot{\theta}^2$ is the familiar *Centripetal contribution*!)

Tangential component: $2\dot{r}\dot{\theta} + r\ddot{\theta}$ $r\ddot{\theta}$ Intuitively clear

$2\dot{r}\dot{\theta}$ is called the *Coriolis* contribution!



Newton's law in plane polar coordinate

$$\vec{F} = m \frac{d\vec{v}}{dt}$$

$$\vec{F} = F_r \hat{r} + F_\theta \hat{\theta} = m[(\ddot{r} - r\dot{\theta}^2)\hat{r} + (2\dot{r}\dot{\theta} + r\ddot{\theta})\hat{\theta}]$$

Newton's law for **radial** direction

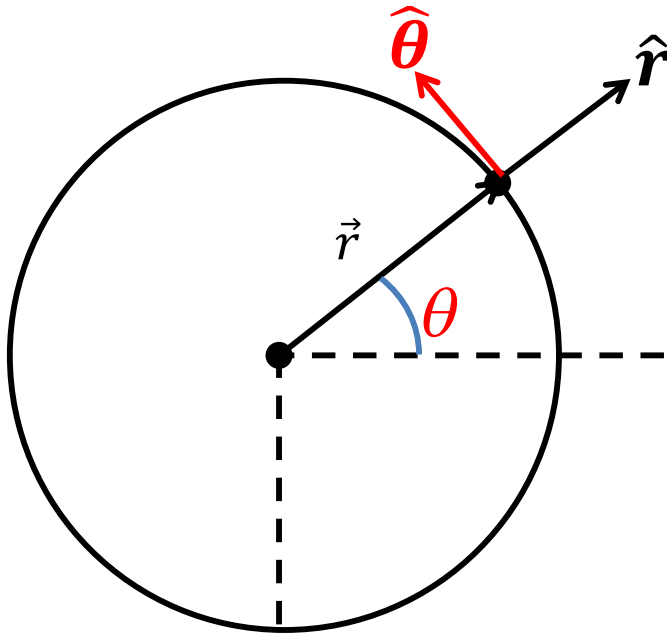
$$F_r = m(\ddot{r} - r\dot{\theta}^2)$$

Newton's law for **tangential** direction

$$F_\theta = m(2\dot{r}\dot{\theta} + r\ddot{\theta})$$



Choice of the coordinate system??



Motion in circular trajectory

Equation of trajectory in polar coordinate

$$r = R = \text{constant}; \theta = \omega t + \frac{1}{2} \alpha t^2$$

The velocity components are

$$v_r = \dot{r} = 0; v_\theta = r\dot{\theta} = R(\omega + \alpha t) = v$$

Acceleration components are

$$a_r = \ddot{r} - r\dot{\theta}^2 = -R(\omega + \alpha t)^2$$

$$a_\theta = r\ddot{\theta} + 2\dot{r}\dot{\theta} = R\alpha = a_t$$

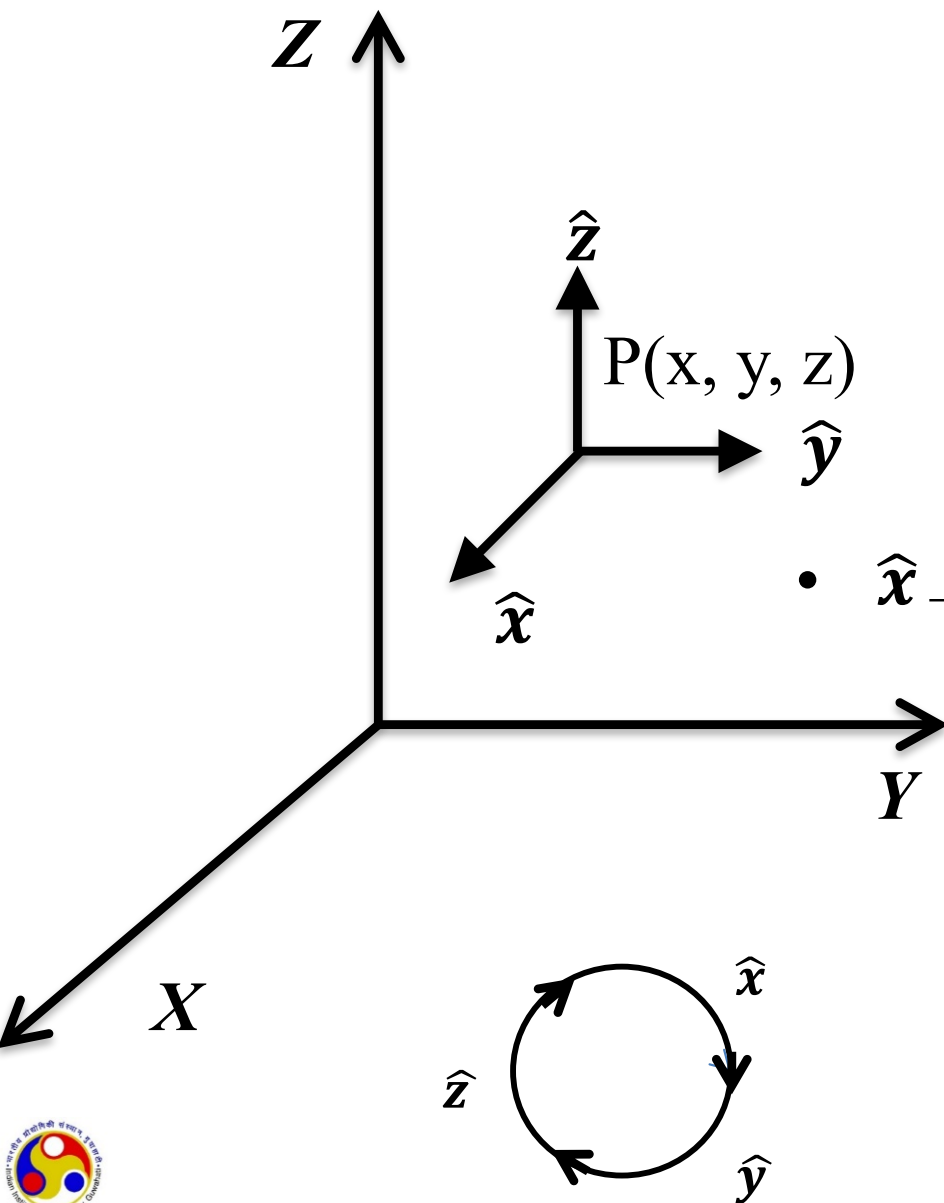
Equation of trajectory in Cartesian coordinate

$$x = R \cos \left(\omega t + \frac{1}{2} \alpha t^2 \right); y = R \sin \left(\omega t + \frac{1}{2} \alpha t^2 \right);$$

Velocity components are

$$v_x = -R(\omega + \alpha t) \sin \left(\omega t + \frac{1}{2} \alpha t^2 \right); v_y = R(\omega + \alpha t) \cos \left(\omega t + \frac{1}{2} \alpha t^2 \right)$$

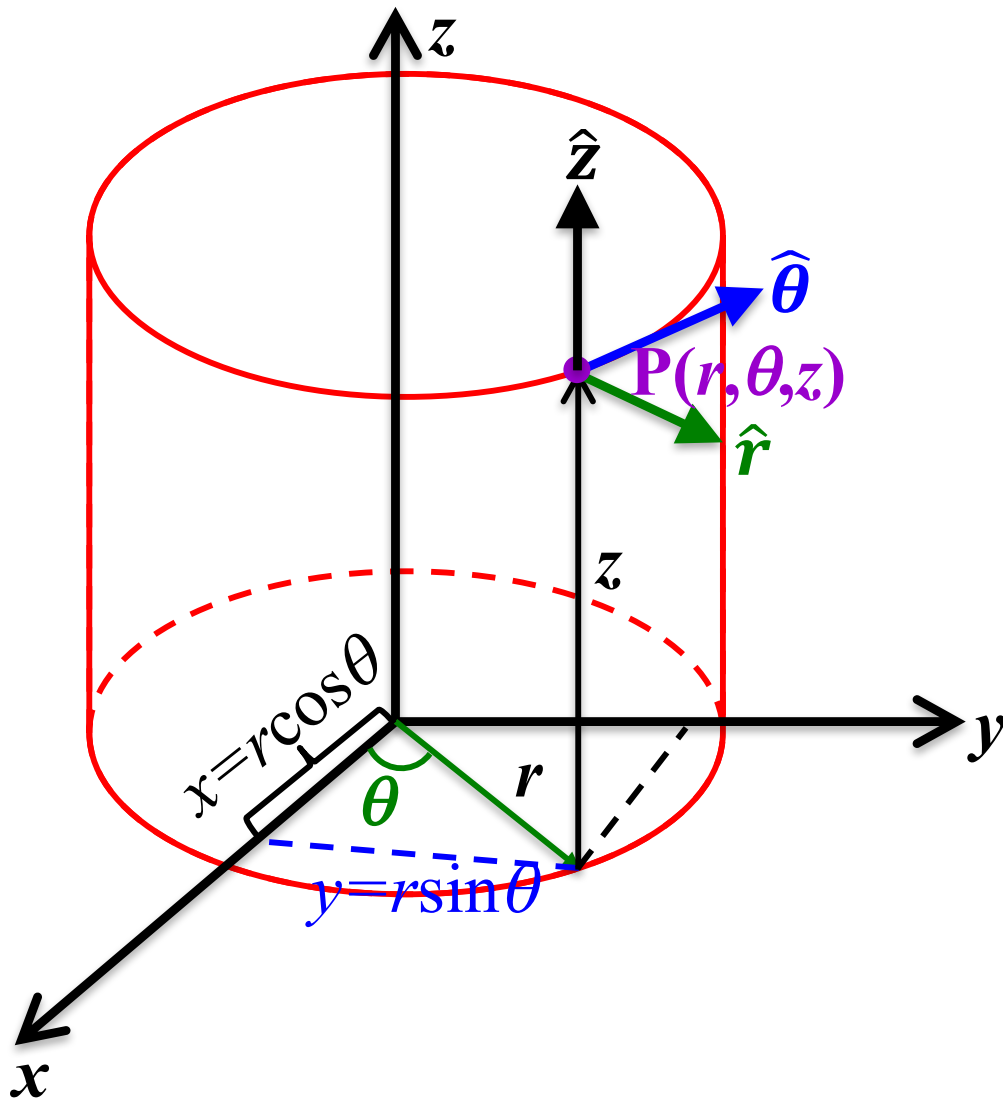
Cartesian co-ordinate system in 3D



- In Cartesian coordinate system, one can define three unit vectors $\hat{x}$, $\hat{y}$ and $\hat{z}$ at every point.
- $\hat{x}$, $\hat{y}$ and $\hat{z}$ are unit vectors in the increasing direction of x, y and z.
- $\hat{x} \perp$ to the YZ plane $\hat{y} \perp$ to the XZ plane and $\hat{z} \perp$ to the XY plane
- $\hat{x}$, $\hat{y}$ and $\hat{z}$ are orthogonal, and directed in the same direction at every points.

$$\hat{x} \times \hat{y} = \hat{z}; \hat{y} \times \hat{z} = \hat{x}; \hat{z} \times \hat{x} = \hat{y};$$

Cylindrical Coordinate System



How to locate a point 'P' in space ?

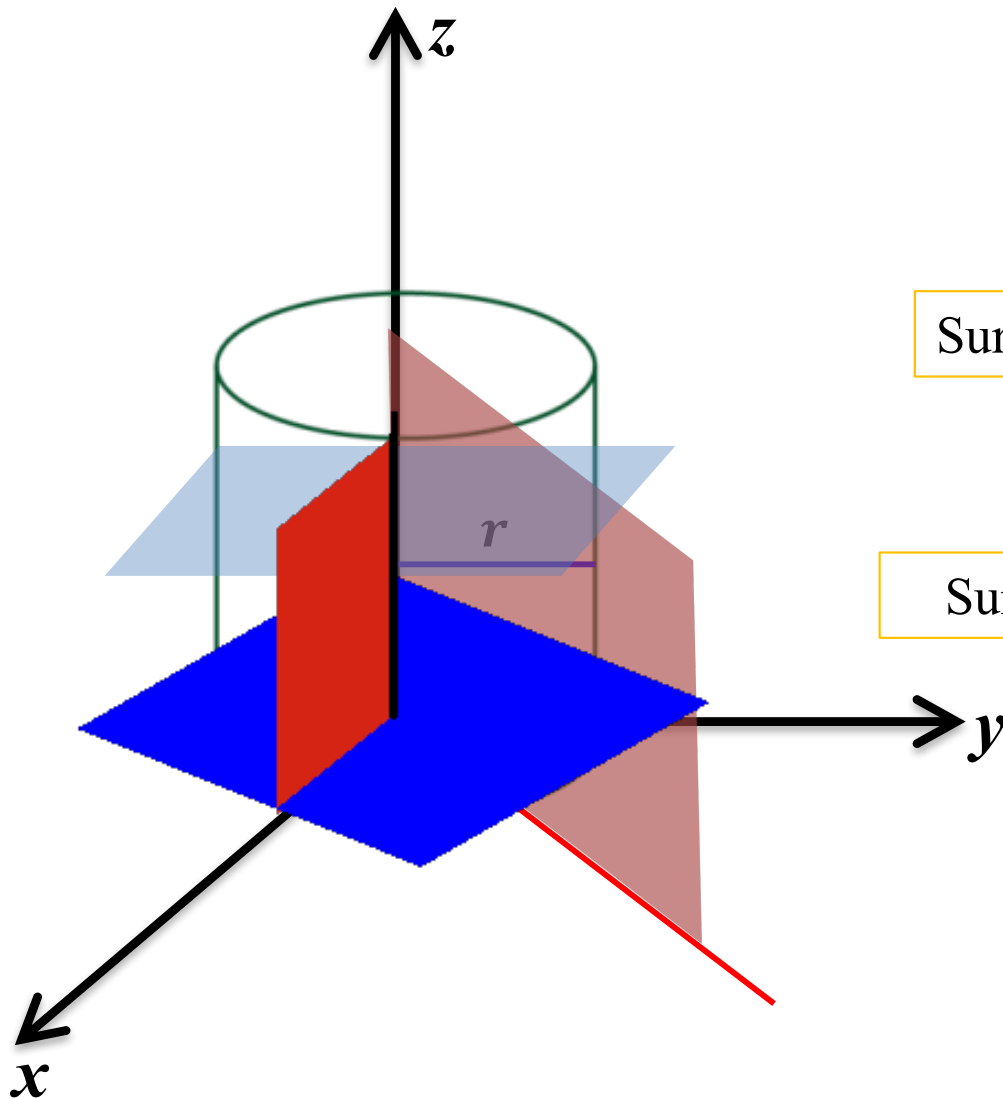
(r, θ, z) coordinates system is known as cylindrical coordinate system

(r, θ) Coordinate of the foot of the point in XY plane

z-Height from the XY plane

$\hat{r}$, $\hat{\theta}$, and $\hat{z}$ are unit vectors along increasing direction of r , θ , and z .

Cylindrical Coordinate System



Note that point (r, θ, z) is at the intersection of three surfaces:

The surface with $r = \text{Constant}$; Cylinder

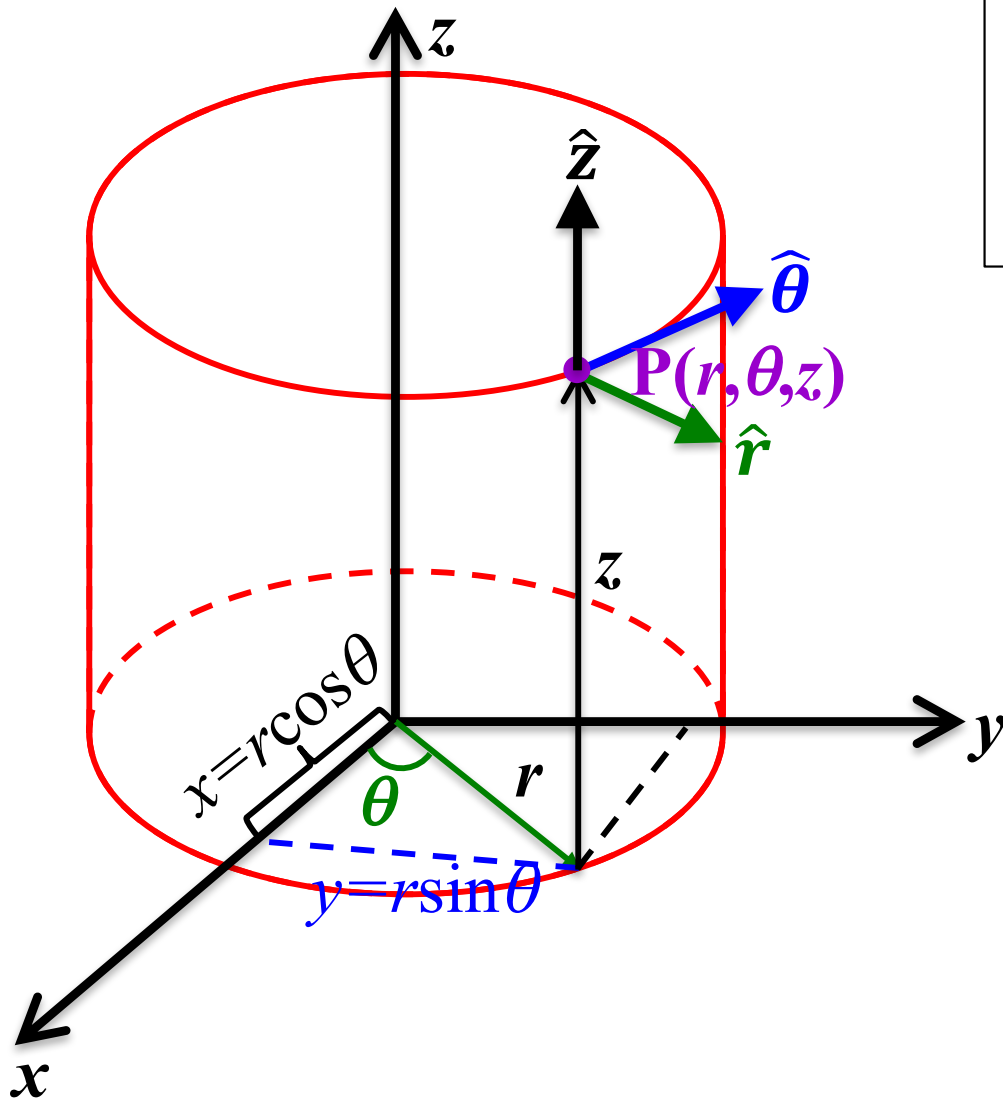
Surface with $\theta = \text{Constant}$; Plane containing Z-axis

Surface $z = \text{Constant}$. Plane Perpendicular Z-axis

The three orthogonal components, r (green), θ (red), and z (blue), each increasing at a constant rate. The point is at the intersection between the three colored surfaces.



Cylindrical Coordinate System



Transformation equation is very similar to polar coordinate with additional z-coordinate.

$$\begin{aligned}x &= r \cos \theta \\y &= r \sin \theta \\z &= z\end{aligned}$$

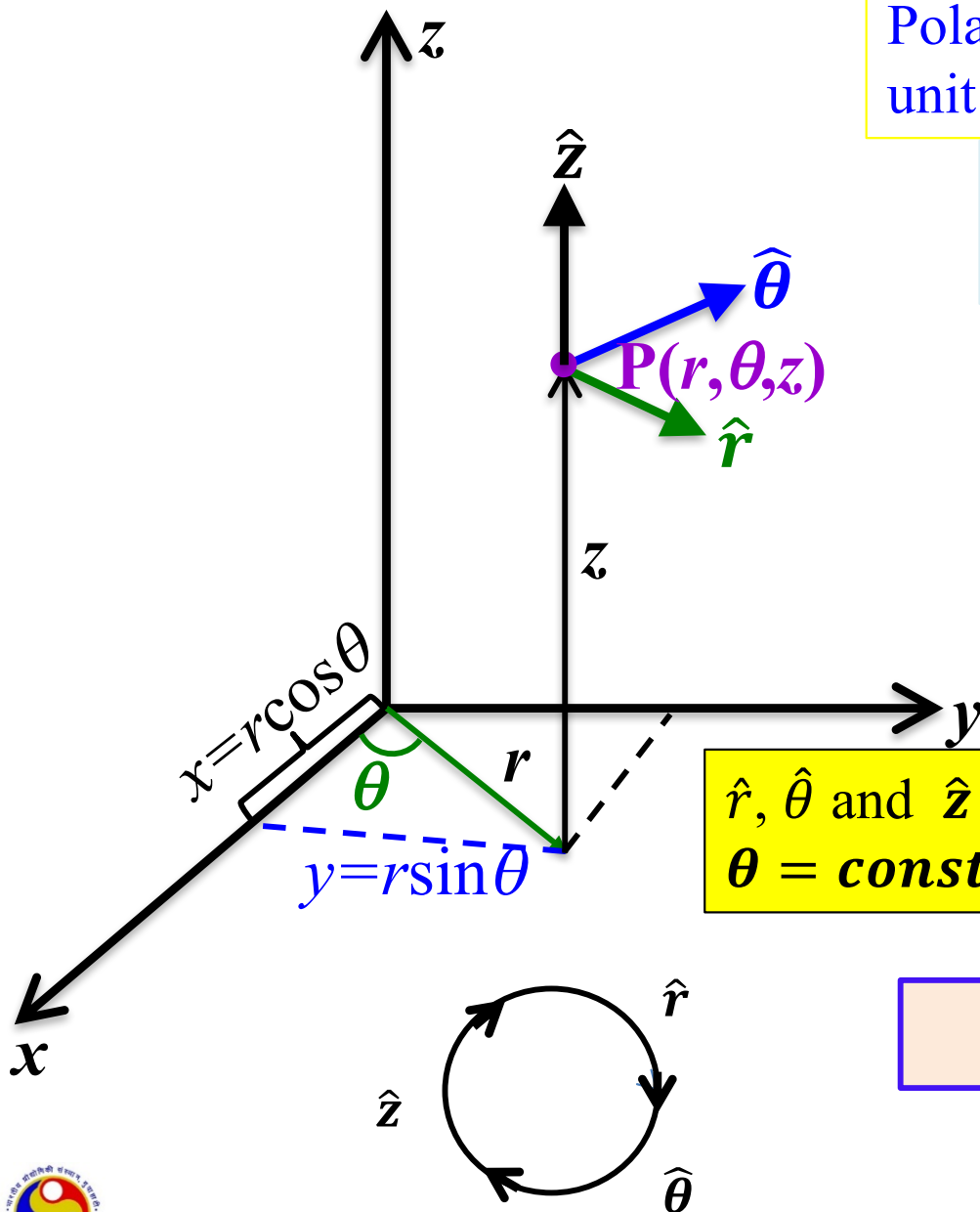
Reverse transformation

$$\begin{aligned}r &= (x^2 + y^2)^{1/2} \\ \theta &= \tan^{-1} \frac{y}{x} \\ z &= z\end{aligned}$$

Note: Instead of (r, θ) , many books use notation (ρ, φ) .



Cylindrical Coordinate System



Polar coordinate unit vectors $(\hat{r}, \hat{\theta})$ + additional unit vector in the z -direction.

$\hat{r}, \hat{\theta}$ and $\hat{z}$ are unit vectors along increasing direction of coordinates r, θ and z .

$$\begin{aligned}\hat{r} &= \cos \theta \hat{x} + \sin \theta \hat{y} \\ \hat{\theta} &= -\sin \theta \hat{x} + \cos \theta \hat{y} \\ \hat{z} &= \hat{z}\end{aligned}$$

$\hat{r}, \hat{\theta}$, and $\hat{z}$ are **orthogonal**, the directions of $\hat{r}, \hat{\theta}$, depend on location but $\hat{z}$ is fixed

$\hat{r}, \hat{\theta}$ and $\hat{z}$ are **perpendicular** to surfaces $r = \text{constant}$; $\theta = \text{constant}$ and $z = \text{constant}$, respectively.

$$\hat{r} \times \hat{\theta} = \hat{z}; \quad \hat{\theta} \times \hat{z} = \hat{r}; \quad \hat{z} \times \hat{r} = \hat{\theta}$$



Position, Velocity, Acceleration, Newton's law in cylindrical coordinate system

Vector components are very similar to polar coordinate + z component

Position vector



$$\vec{r} = r\hat{r} + z\hat{z}$$

Velocity



$$\vec{v} = \dot{r}\hat{r} + r\dot{\theta}\hat{\theta} + \dot{z}\hat{z}$$

$$\frac{d\hat{z}}{dt} = 0$$

Acceleration



$$\vec{a} = (\ddot{r} - r\dot{\theta}^2)\hat{r} + (2\dot{r}\dot{\theta} + r\ddot{\theta})\hat{\theta} + \ddot{z}\hat{z}$$

Newton's law



$$\begin{aligned}\vec{F} &= F_r\hat{r} + F_\theta\hat{\theta} + F_z\hat{z} \\ &= m[(\ddot{r} - r\dot{\theta}^2)\hat{r} + (2\dot{r}\dot{\theta} + r\ddot{\theta})\hat{\theta} + \ddot{z}\hat{z}]\end{aligned}$$



